

Blueprint for a revolutionary German real estate intelligence platform

The German real estate data market has a Zillow-shaped hole — no platform tracks per-property price history, offers cross-portal deduplication, or provides institutional-grade investment analytics. This represents a \$12.7B opportunity (projected German proptech market by 2035) (Market Research Future) that the brick-data platform can capture by combining AI-agent orchestration with KeepaAPI-style historical tracking and hyper-local German data sources. The competitive landscape is fragmented: ImmoScout24 dominates listings but lacks investor analytics, Scoperty failed trying B2C valuations, PropRate offers basic scoring but no historical depth, and GeoMap.immo serves professionals but remains closed-ecosystem. What follows is the complete architectural blueprint — from multi-agent design through database schema to scraping legality — for a solo developer to build this system.

1. The multi-agent architecture that a solo developer can actually ship

LangGraph is the clear framework winner

After evaluating six major frameworks, **LangGraph v1.0** emerges as the production-ready choice for orchestrating this platform's AI agents. With **6.17 million monthly PyPI downloads**, built-in persistence via PostgreSQL checkpointing, and proven deployments at LinkedIn, Replit, and Elastic, it provides the state management and reliability (ZenML) a solo developer needs without operational overhead. The critical insight: pair LangGraph for orchestration with **PydanticAI v1.0** for individual agent definitions — PydanticAI's type-safe structured outputs (Medium) (100% test coverage, (Pydantic) Thoughtworks "Adopt" rating) prevent the #1 production failure in multi-agent systems: malformed agent outputs.

CrewAI prototypes faster but hits a "valley of death" moving to production. (Xcelore) AutoGen/AG2 is being superseded by Microsoft's Agent Framework (still in preview — avoid). OpenAI's Agents SDK lacks built-in state persistence. The industry pattern emerging in 2025-2026 is clear: "prototype with CrewAI, productionize with LangGraph." (ZenML)

Eight specialized agents in a hybrid supervisor-pipeline topology

The optimal architecture uses a **deterministic pipeline** for background data processing and a **LangGraph supervisor** for interactive user queries. This avoids the cost and latency of routing every pipeline step through an LLM while preserving intelligent routing for research queries.

Background pipeline (runs on schedule, no LLM routing needed):

Agent	Type	LLM Usage	Trigger
Scraper Agent	Deterministic (Playwright/Scrapy)	Only for PDF exposé parsing	Celery Beat every 4-6h
Data Enrichment Agent	Hybrid (API calls + LLM for unstructured text)	GPT-4o-mini for text interpretation	Triggered by new listings
Market Analysis Agent	Computational (pandas/numpy) + LLM summaries	GPT-4o-mini for narrative generation	Triggered by enrichment completion
Deal Scoring Agent	Rule-based engine + LLM rationale	GPT-4o-mini for explanation text	Triggered by analysis completion
Notification Agent	Deterministic (no LLM)	None	Triggered by score thresholds

Interactive supervisor (for user-facing requests):

Agent	Type	LLM Usage	Trigger
Research Agent	RAG with pgvector	GPT-4o or Claude Sonnet	User query
Portfolio Agent	Computational + LLM narrative	GPT-4o-mini	User request
Credit Comparison Agent	Scheduled scraper + LLM summary	GPT-4o-mini	Daily + on-demand

The critical design principle: **LLM-last**. Scraping, numerical scoring, notifications, and API-based enrichment are deterministic — no LLM required. Reserve expensive model calls for genuine language understanding tasks. This keeps estimated monthly LLM costs at **\$25-50** for processing ~500 listings/day (90% of calls using GPT-4o-mini at ~\$0.15/1M input tokens).

Background task orchestration with Celery

For a solo developer, **Celery + Redis** provides the simplest production-grade task scheduling. Celery Beat handles cron schedules (scrape ImmoScout24 every 4 hours, check mortgage rates daily at 8am, rescore deals on new data events). The pipeline steps chain naturally: `chain(scrape.s(), enrich.s(), analyze.s(), score.s(), notify.s())`. Redis serves as both message broker and result backend — one dependency, minimal ops.

When pipeline complexity grows (workflows surviving mid-execution crashes, steps taking hours), upgrade to **Temporal** — PydanticAI v1 has built-in Temporal integration for durable agent execution. `Pydantic`

MCP servers provide plug-and-play tool integration

Build five custom **Model Context Protocol** servers as standalone Python files using `mcp.server.fastmcp.FastMCP`. Each exposes tools that agents discover dynamically via `langchain-mcp-adapters`:

1. **Real Estate DB Server** — queries PostgreSQL for listings, scores, historical data
2. **Bodenrichtwert Server** — fetches land values from BORIS-D WMS/WFS APIs
3. **OSM/Transport Server** — queries OpenStreetMap Overpass API for transit connectivity scores
4. **Demographics Server** — queries Zensus 2022 and Destatis GENESIS-Online
5. **Mortgage Rate Server** — serves current rates from scraped data

Adding a new data source means adding one MCP server file — no agent code changes required.

Monitoring, cost control, and error recovery

Langfuse (self-hosted via Docker Compose) beats LangSmith for this project: it's open-source MIT, GDPR-compliant when self-hosted on German infrastructure, and framework-agnostic. Track cost per pipeline run, latency per agent step, error rates, and cache hit rates.

For error recovery: LangGraph's PostgreSQL checkpointing means pipelines resume from the last successful step after crashes. Use `tenacity` for exponential backoff on LLM calls (3 retries max). Implement a dead-letter queue in Celery for failed tasks. Circuit breakers on scrapers — if ImmoScout24 blocks requests, stop retrying and alert via Telegram.

```
python

# Recommended stack summary
FRAMEWORK = "LangGraph v1.0 + PydanticAI v1.0"
TASK_QUEUE = "Celery + Redis"
MONITORING = "Langfuse (self-hosted)"
LLM_DEFAULT = "GPT-4o-mini (90% of calls)"
LLM_RESEARCH = "GPT-4o or Claude Sonnet (10%)"
MCP_TOOLS = "langchain-mcp-adapters"
DEPLOYMENT = "Docker Compose on Hetzner (German DC)"
```

2. A PostgreSQL-native data model built for time-series intelligence

TimescaleDB, pgvector, and PostGIS on a single PostgreSQL instance

The storage architecture runs entirely on **PostgreSQL with three extensions** — no separate databases to manage. **TimescaleDB** converts price history and metrics tables into hypertables with automatic partitioning, compression, and continuous aggregates. `Timescale` **pgvector** enables semantic similarity search ("find listings like this one but cheaper"). `Neon` **PostGIS** handles all geographic queries (radius search, spatial joins with Bodenrichtwert zones). `PostGIS`

This single-database approach is the right call for a solo developer. ClickHouse offers faster analytical queries but adds operational complexity and lacks PostGIS/pgvector equivalents. InfluxDB and QuestDB are purpose-built for metrics, not the mixed relational + time-series + vector + spatial workload this platform requires.

Core schema design spanning 17 tables

The schema separates **properties** (canonical, deduplicated physical locations) from **listings** (individual advertisements from each source). This separation enables cross-portal deduplication — the same apartment listed on ImmoScout24 and Kleinanzeigen resolves to one property record with two linked listings, each maintaining independent price histories.

Prices are stored in cents as BIGINT to avoid floating-point precision issues. The `listing_price_history` table becomes a TimescaleDB hypertable chunked by 7-day intervals, compressed after 30 days, with a 5-year retention policy. At 500 listings scraped daily with 4 observations each, this produces ~730,000 rows per year — well within PostgreSQL's comfort zone, and TimescaleDB's compression typically achieves **5-15x** reduction.

Key tables and their roles:

Table	Purpose	Special Features
<code>properties</code>	Deduplicated physical properties	PostGIS GEOGRAPHY point, GiST spatial index
<code>listings</code>	Per-source listing records	UNIQUE(source_id, external_id), JSONB raw_data
<code>listing_price_history</code>	Every price observation	TimescaleDB hypertable, 7-day chunks, compression
<code>listing_status_history</code>	Immutable event log	Event-sourced lifecycle tracking
<code>area_metrics</code>	PLZ-level aggregates over time	Hypertable, 30-day chunks
<code>credit_offers</code>	Mortgage rate snapshots	Hypertable, segmented by bank
<code>bodenrichtwerte</code>	Official land values	PostGIS MultiPolygon zones, GiST index
<code>deal_scores</code>	Computed 0-100 scores	Component breakdown in JSONB
<code>property_embeddings</code>	Vector embeddings for search	pgvector(1536), HNSW index
<code>alerts</code>	User alert rules	JSONB criteria with GIN index

Continuous aggregates automate the heavy analytics

TimescaleDB continuous aggregates pre-compute four key views that refresh automatically: `Timescale`

- `monthly_price_by_plz` (daily refresh) — powers dashboard price trend charts
- `weekly_price_stats` (hourly refresh) — feeds the alert engine and momentum calculations
- `daily_listing_activity` (hourly refresh) — market barometer showing new/removed listings
- `monthly_mortgage_rates` (daily refresh) — drives yield calculations in deal scoring

These replace manual materialized view maintenance with declarative SQL. A query for "average price per sqm in PLZ 10115 over the last 24 months" hits the pre-computed aggregate instead of scanning millions of raw

price observations.

Listing lifecycle as an event-sourced log

Rather than overwriting listing status, every transition is recorded as an immutable event in `listing_status_history`: `active → price_reduced → reserved → sold → removed → relisted`. The current status is denormalized onto the `listings` table for fast reads, while the full event history supports Keepa-style lifecycle charts. Two helper functions — `fn_record_price_change()` and `fn_record_status_change()` — ensure atomic recording of observations plus status updates.

Semantic search via pgvector enables natural language property discovery

Property embeddings (OpenAI text-embedding-3-small, 1536 dimensions) stored in `property_embeddings` with an HNSW index (m=16, ef_construction=64) support queries like "find me a renovated Altbau with high ceilings near Prenzlauer Berg under €400K." The Research Agent converts natural language to embedding vectors and performs cosine similarity search across the entire listing corpus.

3. Building "Keepa for German Real Estate" — the price tracking engine

Learning from Keepa's data architecture

Keepa stores Amazon price histories as **paired arrays of [timestamp, price]** [Read the docs](#) using a custom epoch (minutes since January 2011) with prices in cents as integers. This compact representation enables tracking billions of price points efficiently. The platform's token-based API system creates economic incentives for efficient querying while generating revenue. [GitHub](#)

The German real estate equivalent stores per-listing data in an identical structure:

```
json
```

```

{
  "listing_id": "IS24-123456789",
  "price_history": [
    [1708000000, 35900000],
    [1710000000, 34500000],
    [1715000000, 33900000]
  ],
  "status_history": [
    [1708000000, "active"],
    [1720000000, "reserved"],
    [1725000000, "sold"]
  ],
  "stats": {
    "days_on_market": 107,
    "price_reductions": 2,
    "total_price_change_pct": -5.6,
    "price_per_sqm": 398824
  }
}

```

CamelCamelCamel's tiered storage reduces costs at scale

CamelCamelCamel's architecture offers a critical lesson: **tiered storage compression**. Hot data (0-7 days) retains per-observation granularity on SSD with 5x compression. Warm data (7-90 days) stores hourly averages on HDD with 15x compression. Cold data (90+ days) aggregates to daily summaries in S3 Parquet with 25x compression. (grokkingthesystemdesign) TimescaleDB's native compression policies automate this — configure (compress_after => INTERVAL '30 days') with (segmentby => listing_id) to achieve similar tiering within PostgreSQL.

The alert engine borrows CamelCamelCamel's inverted index pattern

When a listing's price changes, only alerts registered for that specific listing need evaluation — not all alerts in the system. (grokkingthesystemdesign) An **inverted index** mapping (listing_id → [alert_1, alert_2, ...]) and (criteria_hash → [alert_3, alert_4, ...]) makes evaluation O(1) per listing update. Debounce logic suppresses duplicate notifications within 24-hour windows. A hysteresis buffer requires prices to drop *significantly* below thresholds, preventing spam from penny fluctuations.

Five alert types for real estate investors:

- Price dropped below absolute threshold on a watched listing
- New listing matching configurable criteria (PLZ + rooms + price range + size)
- Area median price crossed a trend threshold
- Listing status changed (sold/removed notification)
- Price reduced by more than X% in a watched area

Chart generation follows Keepa's server-rendered approach

A Chart API endpoint (`GET /api/chart?listing_id=X&range=365&format=png`) generates step-charts server-side using matplotlib — appropriate for price data that holds constant then jumps. Client-side, D3.js or Apache ECharts renders interactive exploration. Per-listing charts show asking price over time with overlaid events. Area-level charts display median price/sqm with confidence bands (25th-75th percentile).

The REST API designed for researchers and developers

```
GET /api/v1/listings/{id}           → Full listing with history
GET /api/v1/listings/{id}/history   → Price + status time series
GET /api/v1/areas/{plz}/metrics     → Current area aggregates
GET /api/v1/areas/{plz}/history     → Historical area metrics
GET /api/v1/search?city=Berlin&rooms=3 → Filtered listing search
GET /api/v1/indices/greix/{city}    → GREIX-linked price indices
GET /api/v1/bodenrichtwerte/{lat}/{lng} → Nearest land value
GET /api/v1/chart/{id}.png          → Generated chart image
GET /api/v1/market/heatmap?bounds=... → Geographic heat data
POST /api/v1/alerts                 → Create price alert
```

Token-based rate limiting (Keepa model) with cursor-based pagination and bulk endpoints supporting up to 1,000 listings per request for researchers. JSON primary format with optional CSV/Parquet for bulk exports.

4. Ten features that make this platform revolutionary

What makes this different from everything in Germany today

No German platform combines historical price tracking + cross-portal deduplication + predictive ML + portfolio analytics + regulatory compliance tools + crowdsourced data. ImmoScout24 dominates listings but offers zero investor analytics. GeoMap.immo serves professionals but remains a closed ecosystem. The opportunity is a **unified intelligence layer** that aggregates fragmented German real estate data into one queryable, analyzable platform.

Feature 1: Micro-market intelligence engine with composite PLZ scoring

Germany has remarkably rich open data that no platform fully aggregates. The micro-market engine combines six data layers into a composite score per postal code area:

Transit connectivity draws from Germany's GTFS.de dataset — one of the world's largest, covering 20,000+ lines and 500,000+ stops. (`Gtfs`) Calculate isochrone-based accessibility scores: how many jobs, schools, and hospitals are reachable within 30 minutes by public transit from each PLZ centroid.

Crime data from BKA Polizeiliche Kriminalstatistik (PKS), available at Kreis level (`BKA`) with some Länder publishing precinct-level data. NRW publishes the most granular data. (`Polizei NRW`) Normalize to burglary rate + property crime rate per 100,000 inhabitants.

Demographics from **Zensus 2022** — Germany's first census to include Nettokaltmiete (net cold rent) and vacancy reasons. (Zensus2022) (Statistisches Bundesamt) Grid-level data at 100m and 1km resolution (CRAN) is freely available under Datenlizenz Deutschland. (Zensus2022) Combine population density, age distribution, income levels, and migration patterns.

Construction permits from Destatis GENESIS-Online API (free with registration). In March 2025, **19,500 new housing permits** were approved (+5.8% YoY). (Statistisches Bundesamt) Track permit trends as a 12-24 month leading indicator of future supply.

School proximity via OpenStreetMap Overpass API for locations, supplemented by crowdsourced Google Maps reviews as a quality proxy (no unified German school rating system exists — a gap itself).

Gentrification probability using a composite ML model trained on: new café/coworking space openings (OSM/Google Maps data), Google Trends volume for neighborhood names, Airbnb listing density changes, (Eagle Alpha) rent gap analysis, and demographic age-shift patterns.

Feature 2: XGBoost-powered price prediction with SHAP explainability

Research on **2.4 million German rental observations** confirms XGBoost as the consistently top-performing model for German real estate, (FUPress) reducing forecasting errors by **68%** versus simple regression.

(CoreCast Blog) A 2025 paper demonstrates a two-stage approach — cluster by location features first, then build property-specific models per cluster — improving MAE by up to **58%**. (arXiv)

Key features ranked by SHAP importance: living area (sqm), location (PLZ, lat/lng, Wohnlage classification), construction year, number of rooms, energy efficiency (kWh/m²a), property condition, floor level, building type, and parking availability. Adding alternative data (transit scores, POI density, crime rates) significantly improves accuracy beyond traditional variables. (Propri)

The platform should produce not just point estimates but **confidence intervals** — inspired by Zillow's Neural Zestimate which uses quantile regression to output ranges. (zillow) Display: "Estimated value: €285,000 ± €15,000 (confidence: 85%)."

Feature 3: "Buy signal" alerts using cross-domain alternative data

The most revolutionary feature fuses real estate data with non-traditional signals. McKinsey research confirms that when traditional variables (income, vacancy, build year) predict values at ~40% accuracy, adding alternative data sources raises this substantially:

- **Infrastructure impact:** Studies show **10-25% property value increases** within 500m of new transit stations. Track Bundesverkehrswegeplan announcements and municipal infrastructure plans.
- **Google Trends:** Rising search volume for neighborhood names correlates with gentrification and price appreciation 6-18 months ahead.
- **Bundesagentur für Arbeit:** Employment data by Kreis — direct housing demand driver. Rising job vacancy rates signal incoming population growth.
- **Handelsregister:** Company registrations/dissolutions by location serve as an economic activity proxy.

- **Business type changes:** The transition from Spätkauf to specialty coffee shop, documented via OSM/Google Maps POI changes, predicts neighborhood appreciation. (DataDrivenInvestor)

Aggregate these signals into a color-coded system inspired by Hopper's flight price prediction: (StayFi) ● **Good time to buy** (underpriced + appreciating area) / ● **Hold** / ● **Wait** (overpriced or declining signals).

Feature 4: Mietpreisbremse compliance checker

Germany's rent brake law caps net cold rent at the ortsübliche Vergleichsmiete + 10% (CONNY) in **410 of 11,000 municipalities** designated as tight housing markets. (IamExpat) No major platform offers automated compliance checking. The feature: input an address + proposed rent → auto-lookup applicable Mietspiegel (Berlin, Munich, Hamburg all publish qualified versions) → calculate maximum legal rent → flag violations → check exemptions (post-2014 first-use, comprehensive modernization).

An open-source starting point exists: the **mietenwatch/mw-mietspiegel-api** GitHub project provides a Berlin Mietspiegel API calculating Wohnlage and comparative rent per address. (GitHub)

Feature 5: Zwangsversteigerungen (foreclosure auction) aggregator with AI analysis

Court auction data from **475+ Amtsgerichte** is published on zvg-portal.de (Justizportal) but fragmented and difficult to analyze. The platform aggregates all auctions, compares Verkehrswert (appraised value) against estimated market value from the ML model, calculates potential discount percentages, and alerts investors on properties matching their criteria. **ZvgScout.com** demonstrates demand for this feature with AI-powered Gutachten (appraisal) analysis and map views. (ZvgScout)

Features 6-10 in brief

Portfolio analytics with German tax engine — Track AfA depreciation (2% linear post-1925, 5% degressive for new builds since 2023), Grunderwerbsteuer by state (3.5% in Bavaria to 6.5% in Brandenburg/NRW), and Spekulationsfrist countdown (10-year holding period for tax-free capital gains). (Financemate) No German platform offers this.

Automated comp analysis — Select comparable properties using PLZ proximity, size ($\pm 15\%$), rooms, Baujahr, Wohnlage, and Energieausweis class. (Oreate AI) Apply German-specific adjustments for floor level, Altbau premium, and elevator presence. Score confidence based on comp availability and attribute similarity.

Interactive multi-layer heat maps — Choropleth maps showing price trends, rental yields, supply/demand ratios, days-on-market, demographics, and energy efficiency — all togglable on one map. Base layer uses **Zensus 2022 1km² grid data** (Nrw) for demographic overlays.

Crowdsourced renovation cost index — No systematic German data exists on actual renovation costs by category and region. Let investors anonymously submit costs (kitchen: €8-25K, bathroom: €5-15K, energy retrofit: varies). Aggregate into a PLZ-level index. Research shows energy efficiency improvements of 100 kWh/m²a correlate with **~6.9% price increases**, stronger in rural areas. (ScienceDirect)

Market cycle dashboard — Position each German city on the Mueller 4-phase cycle model (Recovery → Expansion → Hypersupply → Recession) using vacancy rates, rental growth vs. inflation, permit activity trends, and affordability indices. Cycles span **7-9 years** (85-110 months). (Cycles)

5. Scraping German real estate portals — legal framework and technical strategy

The legal landscape permits careful scraping but demands GDPR caution

A **BGH landmark decision from November 2024** (VI ZR 10/24) ruled that mere loss of control over personal data constitutes immaterial damage under GDPR Art. 82(1), with indicated compensation of ~€100 per person. This makes GDPR compliance non-negotiable: **strip personal data** (agent names, phone numbers, emails) immediately during parsing unless strictly necessary.

However, scraping factual real estate data is legally defensible under three principles. First, § **44b UrhG** (Germany's TDM exception since 2021) permits commercial text and data mining of publicly accessible data, provided rights holders haven't opted out via machine-readable means (robots.txt). Second, the **Datenbankherstellerrecht** (database sui generis right) protects only against extracting "substantial parts" (~20%+) of a database — individual listings are "insubstantial parts" and freely usable. (Aufrecht) Third, **factual data (prices, sizes, addresses) lacks Schöpfungshöhe** (creative threshold) and is not individually copyrighted, (Rechtzweignull) though listing descriptions and photos ARE protected.

Practical legal rules: Only scrape publicly accessible pages (no login walls). Respect robots.txt.

(Forschung & Lehre) Never copy >20% of any single database. (Uni-stuttgart) Don't republish copyrighted descriptions or images — extract only structured factual data. Rate-limit requests. Document your legitimate interest under GDPR Art. 6(1)(f). Engage a German IT lawyer before production deployment.

ImmoScout24 requires either API partnership or sophisticated stealth

ImmoScout24 offers a **comprehensive REST API** at (api.immobilienscout24.de) (immobilienscout24) with OAuth 1.0 authentication, (ImmobilienScout24) (ImmobilienScout24) including Search, Expose, Geo Autocomplete, and Energy Certificate endpoints. However, bulk search across all marketplace listings requires a **paid data partnership** — the API is designed primarily for agents managing their own listings.

Without partnership access, ImmoScout24 employs aggressive bot detection: (Scrapfly) (HasData) **Reese84** **cookie-based behavioral analysis**, GeeTest CAPTCHAs, IP-based rate limiting, and browser fingerprinting that detects headless browsers. The most active open-source project, Flathunter (2K+ GitHub stars), reports that ImmoScout24 **cannot be crawled without Capmonster** (CAPTCHA-solving service), and even then requires periodic manual cookie refresh.

Recommended approach priority:

1. **Apply for official API partnership** — provides legal, structured access
2. **Reverse-engineer embedded JSON** — search pages embed listing data in (<script>) tags parseable via XPath
3. **Playwright with residential proxies + Capmonster** — last resort, throttled to 1 request per 5-10 seconds

Kleinanzeigen.de uses Akamai Bot Manager — the hardest to scrape

Kleinanzeigen.de (formerly eBay Kleinanzeigen) has **no official API** and deploys Akamai Bot Manager with sophisticated device fingerprinting, plus DataDome real-time detection, geo-blocking requiring German residential IPs, and dynamic CSS selectors that break static scrapers. Best approach: use managed scraping APIs (ScrapingBee, Scrappey) that handle anti-bot bypass, or Playwright with German residential proxies and persistent browser sessions. [RapidSeedbox](#)

Open data sources eliminate scraping for critical enrichment data

Seven high-value datasets require no scraping at all:

- **BORIS-D** (Bodenrichtwerte): WMS/WFS services with zone boundaries and land values per sqm. [Financemate +2](#) NRW provides historical Shapefiles back to 2011 [Open.NRW](#) under Datenlizenz Deutschland Zero. [Nrw](#) Brandenburg offers an OGC API.
- **GREIX** (German Real Estate Index): The only German index based on **actual transaction prices** (2M+ notarized records from Gutachterausschüsse), covering 24 cities back to 1960. [Econtribute](#) Freely accessible at greix.de. [Econtribute](#)
- **Zensus 2022**: Grid data at 100m-1km resolution with Nettokaltmiete, vacancy rates, building age, and heating energy sources. Free REST API available. [Zensus2022](#) [German Federal Statistical Office](#)
- **Destatis GENESIS-Online**: Building permits, housing completions, price indices via REST API. [restatis](#)
- **Deutsche Bundesbank**: SDMX REST API at [api.statistiken.bundesbank.de](#) for residential property price indices.
- **GTFS.de**: Complete German public transit dataset — 20,000+ lines, 500,000+ stops. [Gtfs](#)
- **OpenStreetMap**: Excellent German coverage of building footprints, amenities, and transit via Overpass API.

Cross-source deduplication uses probabilistic entity resolution

The same apartment listed on ImmoScout24 and Kleinanzeigen at slightly different prices must resolve to one canonical property. The pipeline: normalize German addresses [Esri](#) (Str. → Straße, standardize umlauts), geocode via Nominatim to canonical lat/lng, block candidate pairs by PLZ + room count + approximate price range ($\pm 15\%$), then compute similarity scores across address (Jaro-Winkler), coordinates (Haversine $< 100\text{m}$), price (within 5%), size (within 10%), and optionally image perceptual hashing.

Python's **Splink** or **recordlinkage** libraries implement Fellegi-Sunter probabilistic matching. [Science](#) [Medium](#)
Threshold: combined score > 0.85 → auto-merge; $0.6-0.85$ → manual review; < 0.6 → distinct properties.
Connected components clustering [Medium](#) groups all matched pairs into canonical property entities. [Medium](#)

6. Where competitors fall short and the market gap stands wide open

Germany's real estate data landscape is a decade behind the US and Australia

The comparison is stark. Zillow offers automated valuations for **104 million homes**, [Zillow](#) property-level value histories going back years, [zillow](#) freely downloadable ZIP-code-level research data, and a commercial valuation API. Redfin updates listings every 5 minutes from MLS and publishes housing data for researchers. Australia's Domain.com.au covers 95% of homes with automated valuations [Pricefinder](#) via its Pricefinder subsidiary.

Germany has none of this at consumer or prosumer grade. The **1,318 active German proptech companies** (H1 2025 record) are overwhelmingly focused on energy efficiency (81% of proptech funding goes to Enpal, Zolar, Sunvigo) or transaction facilitation (McMakler, Homeday). Investment analytics is dramatically underfunded.

The lesson from Scoperty's failure — and why B2B wins

Scoperty attempted to publicly map ~35 million German property valuations but was liquidated in late 2022 after failing its financing round. The cause: **cultural resistance** to US-style transparent property valuations, criticism of data accuracy, and privacy backlash from property owners. The lesson is not that property intelligence fails in Germany — it's that **B2C public valuations** face cultural friction. A B2B/prosumer platform targeting investors, family offices, and asset managers avoids this entirely.

Five critical gaps no German platform fills

No per-property historical price tracking — The closest is GREIX for transaction-based indices at city/neighborhood level, [Econtribute](#) [University of Bonn](#) but no platform lets you see "this apartment was listed at €359K on Feb 15, reduced to €345K on March 10, then to €339K on June 1." This KeepaAPI-style tracking is the single most differentiating feature.

No cross-portal intelligence — The same property appears on ImmoScout24, Immowelt, Kleinanzeigen, and bank portals simultaneously. No tool reliably deduplicates across portals or tracks which sources have the best price. PropRate's meta-search aggregates but deduplication quality is unclear.

No unified open data layer — Bodenrichtwerte, Gutachterausschuss data, building permits, demographics, transit data, and Mietspiegel all exist in separate systems with different formats and access methods. Nobody combines them into one queryable intelligence layer at PLZ resolution.

No portfolio analytics with German tax modeling — No platform calculates aggregate cashflow projections across multiple properties with AfA depreciation schedules, [Financemate](#) Grunderwerbsteuer variations by state, or Spekulationsfrist countdown timers.

No ML-powered deal scoring with alternative data — PropRate offers a 100-factor scoring system but uses only traditional real estate metrics. Nobody integrates Google Trends, business opening data, infrastructure announcements, or employment trends into automated buy/hold/wait signals.

The existing repo and what comes next

The brick-data repository at github.com/hirohamada1/brick-data appears to be private or recently created — it couldn't be accessed via public search or direct URL fetching. This suggests the project is at early-MVP stage,

likely with initial scraping infrastructure and basic data storage. The architecture described in this report can serve as the complete technical specification for evolving that codebase into a production system.

7. The recommended technology stack and implementation roadmap

Complete stack for a solo developer on Hetzner

Backend:	FastAPI (Python 3.12+)
Agent Framework:	LangGraph v1.0 + PydanticAI v1.0
Task Queue:	Celery + Redis
Database:	PostgreSQL 16 + TimescaleDB + pgvector + PostGIS
Cache:	Redis (shared with Celery broker)
Scraping:	Playwright + Scrapy (scrapy-playwright)
Proxies:	Bright Data residential German IPs
Monitoring:	Langfuse (self-hosted) + Sentry
LLMs:	GPT-4o-mini (default) + GPT-4o (research queries)
MCP:	5 custom FastMCP servers
Frontend:	Next.js 14+ (SSR for SEO)
Maps:	MapLibre GL JS + PMTiles
Charts:	Apache ECharts (client) + matplotlib (server API)
Infrastructure:	Docker Compose on Hetzner Cloud (German DC, GDPR)

Four-phase roadmap

Phase 1 (Weeks 1-3): Core data pipeline. Playwright scraper for ImmoScout24 search results → PostgreSQL with TimescaleDB → basic price change detection → Telegram notifications for new listings matching criteria. No LLM required. This delivers immediate value.

Phase 2 (Weeks 4-6): Intelligence layer. LangGraph pipeline orchestration with checkpointing. BORIS-D and Zensus 2022 data integration. Rule-based deal scoring (yield calculation, price vs. Bodenrichtwert, days-on-market). Celery Beat scheduling. Langfuse monitoring.

Phase 3 (Weeks 7-10): User-facing platform. FastAPI REST endpoints. Next.js frontend with interactive price history charts and heat maps. Research Agent with RAG (pgvector). Cross-source deduplication pipeline. Alert configuration UI.

Phase 4 (Weeks 11+): Revolutionary features. XGBoost price prediction model trained on accumulated data. Portfolio analytics with German tax engine. Zwangsversteigerung aggregation. Credit comparison agent. Market cycle dashboard. Crowdsourced renovation costs.

Conclusion: three insights that change how to approach this project

First, **the agent architecture should be LLM-last, not LLM-first**. The instinct to run everything through AI

agents is wrong for this use case. Scraping, numerical scoring, alerts, and data enrichment via APIs are deterministic tasks that work better as traditional Python code orchestrated by LangGraph's state machine. Reserve LLM calls for genuine language tasks: parsing unstructured exposé PDFs, generating market narratives, and powering the Research Agent's natural language interface. This cuts monthly LLM costs from hundreds of dollars to under \$50.

Second, **Germany's fragmented data landscape is both the biggest challenge and the deepest moat.** The US has centralized MLS, public transaction records, and Zillow's decade head start. Germany has Bodenrichtwerte in separate WFS services per state, Gutachterausschuss data that varies by municipality, Mietspiegel in different formats per city, and building permits buried in GENESIS-Online API tables. The platform that unifies these sources into one coherent intelligence layer creates an extraordinarily defensible data advantage — because the integration work is the product.

Third, **the Scoperty lesson points toward B2B, not B2C, but the real opportunity is the "prosumer" investor.** Institutional platforms like GeoMap.immo and 21st Real Estate serve professionals at enterprise prices. Consumer platforms like ImmoScout24 serve homebuyers with zero analytics. The gap is the **investor managing 2-50 units** — sophisticated enough to need yield calculations, tax modeling, and market intelligence, but underserved by both consumer and enterprise tools. This is the market position where "Keepa for German Real Estate" becomes indispensable.